

TIANLE DUAN

+1 (942) 688-0915 | harry.duan@mail.utoronto.ca

linkedin.com/in/tianle-duan | github.com/dtl915 | leetcode.com/u/HarryDuan

EDUCATION

University of Toronto, St. George Campus

Honours Bachelor of Science – Computer Science Specialist

- **GPA:** 3.92/4.0 (First Year)
- **Planned Focuses:** Artificial Intelligence; Web and Internet Technologies
- **Relevant Coursework:** CSC110, CSC111, CSC240 (Enriched Intro to Theory of Computation), MAT157 (Analysis I – Specialist), MAT223

Toronto, ON

Sept 2025 – May 2029

AWARDS & ACHIEVEMENTS

USACO Silver Division | Competitive Programming

Jan 2024

- Solved algorithmic problems involving binary search, greedy algorithms, prefix sums, two pointers, and graph traversal (BFS/DFS) under time constraints using C++

LeetCode Contest Rating: 1850+ | Top ~5% Globally

Current

- Solved 100+ algorithm problems across arrays, graphs, dynamic programming, and other topics

EXPERIENCE

Advanced Systems Development Co., Ltd. (ASDC)

Beijing, China

Software Engineering Intern

Jun 2026 – Jul 2026

- Developed an automated entity and relation extraction pipeline for multimodal archival documents, integrating **Whisper**, **PaddleOCR**, **EasyOFD**, and **PyMuPDF** for document processing and knowledge graph construction
- Implemented LLM-based information extraction workflows using cloud-hosted **Qwen3.5** through **OpenAI-compatible APIs**, incorporating **LangChain** recursive summarization and **RDF/JSON-LD** serialization to generate structured knowledge graph data
- Built and deployed a full-stack knowledge graph visualization platform on the archives' internal network, developing **FastAPI** backend services and a **React + TypeScript** frontend with **vis-network** for interactive **Neo4j** graph exploration

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL, Cypher

Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Competitive Programming

AI / LLM: LLM Integration (OpenAI-compatible APIs), OCR (PaddleOCR), Speech-to-Text (Whisper), Knowledge Graphs

Web Development: FastAPI, Flask, React

Databases: Neo4j, MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Docker, IBM Cloud (Code Engine), Linux

PROJECTS

StoryForge | GenAI Genesis Hackathon – 4-person team

Mar 2026

- Built an AI-driven interactive-fiction platform on a **multi-agent LLM architecture** using **GPT-4o** and **IBM watsonx.ai** (Granite model for content-safety filtering)
- Proposed a “director” narrator agent to coordinate character agents and drive the storyline in the system design
- Deployed with **Docker** and **IBM Code Engine** (serverless), storing images in **IBM Container Registry** and hosting static frontend assets on **Cloud Object Storage**

Flashcard Learning App | Python, Tkinter, SQLAlchemy, MySQL

Sep 2024 – Mar 2025

- Developed a desktop application for vocabulary learning with a **Tkinter GUI**
- Implemented **XML parsing** to import words from YouDao Dictionary exports
- Designed database schema using **SQLAlchemy ORM** with **MySQL** for persistent storage
- Built features for grouping words and reviewing them as flashcards

Weather Lookup App | Python, Flask, BeautifulSoup, MySQL

Jul 2025

- Built a web application to **search historical weather data** for any city and date
- Implemented **web scraping** using BeautifulSoup to extract data from external weather source
- Designed a **MySQL caching layer** to store scraped data for faster subsequent queries